

UNIVERSITI KEBANGSAAN MALAYSIA
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Device Physics for Nanoscale CMOS: A Working Revision

KKKC6353 Nanoelectronic Devices — Lecture 1, Document 2 of 3

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How to use this document

This is a **revision**, not an introduction. It assumes you have met semiconductor physics before and is written to re-establish the specific results this module depends on, in one consistent notation, as quickly as is consistent with being correct.

It is organised in three movements:

- **Sections 1 and 2** compress bulk carrier physics and transport. Read these fast. If any of it is new rather than familiar, that is a signal to go back to a standard text (Sze, or Taur and Ning) before Lecture 2.
- **Section 3 builds the MOSFET.** It is not optional: everything in Document 3 is a perturbation on the device constructed here. Section 3.5, on subthreshold conduction, contains the single most important result in the module.
- **Section 4 is where this document earns its place.** It identifies exactly where the classical picture built in Sections 1–3 stops being valid at modern dimensions, and why. This is the boundary between an undergraduate device course and this one.

Boxes marked **Common pitfall** flag explanations that appear in widely used texts and lecture notes but do not survive scrutiny. They are included deliberately: you should know which received explanations are wrong, and why.

Notation

This document fixes the following conventions and uses them without exception. Take particular note of the first five rows: many texts use V_T for the thermal voltage, and E for both field and energy.

Symbol	Meaning	Value / note
$\mathcal{E}$	electric field	V/cm. Never E — that is reserved for energy
E_C, E_V, E_i, E_F	band edges, intrinsic level, Fermi level	eV
E_g	bandgap	1.12 eV for Si at 300 K
U_T	**thermal voltage** kT/q	25.9 mV at 300 K
V_T	**threshold voltage**	—
ψ_s	surface potential	V
ϕ_F	bulk Fermi potential, $U_T \ln(N_A/n_i)$	V
m_n^*, m_p^*	effective mass of electrons, holes	kg, or in units of m_0
μ_n, μ_p	**bulk** mobility	$\mathrm{cm^2 V^{-1} s^{-1}}$
μ_{eff}	**effective inversion-layer** mobility	not the same quantity — see Section 4.3
τ	mean free time between collisions	s
λ	mean free path	cm
C_{ox}	gate oxide capacitance per unit area	$\mathrm{F/cm^2}$
C_d	depletion capacitance per unit area	$\mathrm{F/cm^2}$

Symbol	Meaning	Value / note
L	**physical** gate length	not the node name

1 Carriers in silicon

1.1 Bands and effective mass

A periodic crystal potential turns the free-electron parabola into bands. Near a band extremum the dispersion is again approximately parabolic,

$$E(k) \;\approx\; E_C + \frac{\hbar^2 k^2}{2m^*},$$

which lets us keep Newtonian equations of motion provided we replace the free electron mass m_0 by an **effective mass** m^* that absorbs the entire influence of the lattice. Formally $1/m^* = \hbar^{-2}\,\partial^2 E/\,\partial k^2$: *effective mass is inversely proportional to band curvature.* A flat band means a heavy carrier.

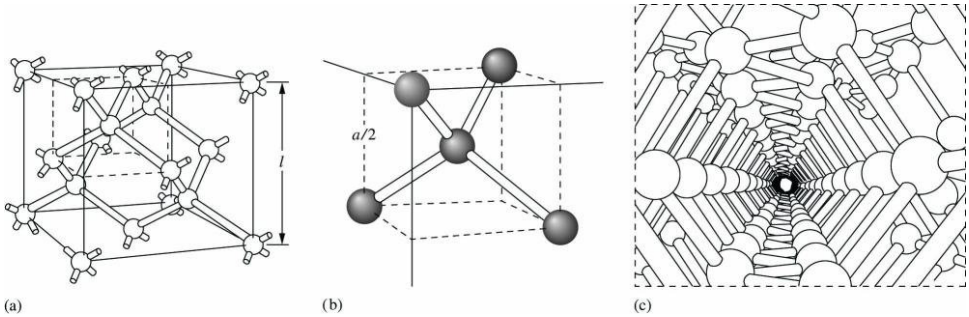


Figure 2.1. *The silicon crystal: (a) the diamond-lattice unit cell; (b) each atom covalently bonded to four nearest neighbours; (c) the lattice viewed along a crystal axis. Textbook figure © McGraw-Hill; reproduced from S. Levkov, NJIT ECE 271 lecture slides, Ch. 2.*

Almost every comparative statement in this module — why one material is faster than another, why strain helps, why holes are slower — reduces to effective mass.

Silicon's conduction band has six equivalent minima along ⟨100⟩, each an ellipsoid with longitudinal mass $m_l = 0.92\,m_0$ and transverse mass $m_t = 0.19\,m_0$. Two different averages are needed depending on the question:

	Conduction band	Valence band
Conductivity effective mass (for transport)	$m_c^* \approx 0.26\,m_0$	$\approx 0.39\,m_0$
Density-of-states effective mass (for carrier counting)	$m_d^* \approx 1.08\,m_0$	$\approx 0.81\,m_0$

For comparison, GaAs has a single, far more curved conduction-band minimum at Γ with $m^* \approx 0.063\,m_0$ — four times lighter than silicon, which is the whole reason III–V channels are of interest.

> **Common pitfall.** Introductory texts sometimes explain $\mu_p < \mu_n$ by saying that hole conduction requires "multiple covalent bond disruptions" while electrons "move freely". That is not a mechanism. The correct statement is that the valence band is **flatter** than the conduction band, so $m_p^* > m_n^*$, so, from $\mu = q\tau/m^*$, $\mu_p < \mu_n$. Nothing to do with bonds.

1.2 Intrinsic carriers and doping

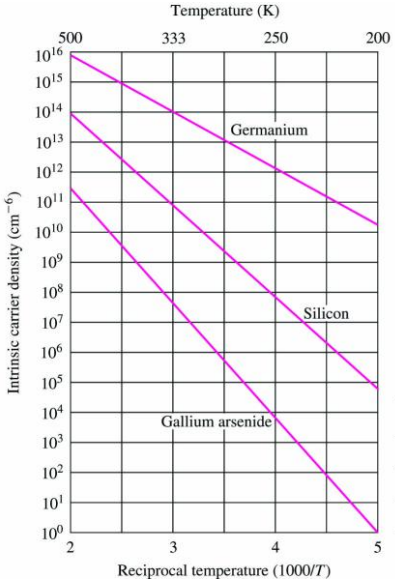
Carrier densities follow from the density of states and the occupation function. Under Boltzmann statistics (valid when E_F sits at least a few kT inside the gap):

$$n = N_C \exp\left(-\frac{E_C - E_F}{kT}\right), \qquad p = N_V \exp\left(-\frac{E_F - E_V}{kT}\right),$$

with $N_C \approx 2.8 \times 10^{19}\ \mathrm{cm^{-3}}$ and $N_V \approx 1.04 \times 10^{19}\ \mathrm{cm^{-3}}$ for Si at 300 K. Multiplying removes E_F and gives the **mass-action law**:

$$np = N_C N_V \exp\left(-\frac{E_g}{kT}\right) = n_i^2.$$

For silicon at 300 K the currently accepted value is $n_i \approx 9.65 \times 10^9\ \mathrm{cm^{-3}}$. Older texts quote 1.45×10^{10} or 1.5×10^{10}. The lower figure is the reassessment by Altermatt *et al.* (2003), which corrected earlier measurements for bandgap narrowing; it is the one to use. The value matters, because ϕ_F — and hence V_T — depends on it logarithmically.



	B ($\mathrm{K^{-3} \cdot cm^{-6}}$)	E_G (eV)
Si	1.08×10^{31}	1.12
Ge	2.31×10^{30}	0.66
GaAs	1.27×10^{29}	1.42

Figure 2.2. *Intrinsic carrier density against inverse temperature for germanium, silicon and gallium arsenide. The slope of each line is set by the bandgap, which is why n_i in Ge exceeds that in GaAs by about six orders of magnitude at room temperature. Textbook figure © McGraw-Hill; reproduced from S. Levkov, NJIT ECE 271 lecture slides, Ch. 2.*

Charge neutrality in a doped, fully ionised sample requires

$$p + N_D^+ - n - N_A^- = 0,$$

which with $np = n_i^2$ gives, for n-type material with $N_D - N_A \gg n_i$, simply $n \approx N_D - N_A$ and $p \approx n_i^2 / n$.

	IIIA	IVA	VA	VIA
	5 10.811 **B** Boron	6 12.01115 **C** Carbon	7 14.0067 **N** Nitrogen	8 15.9994 **O** Oxygen
	13 26.9815 **Al** Aluminum	14 28.086 **Si** Silicon	15 30.9738 **P** Phosphorus	16 32.064 **S** Sulfur
IIB				
30 65.37 **Zn** Zinc	31 69.72 **Ga** Gallium	32 72.59 **Ge** Germanium	33 74.922 **As** Arsenic	34 78.96 **Se** Selenium
48 112.40 **Cd** Cadmium	49 114.82 **In** Indium	50 118.69 **Sn** Tin	51 121.75 **Sb** Antimony	52 127.60 **Te** Tellurium
80 200.59 **Hg** Mercury	81 204.37 **Tl** Thallium	82 207.19 **Pb** Lead	83 208.980 **Bi** Bismuth	84 (210) **Po** Polonium

Figure 2.3. *The part of the periodic table that matters for silicon: group III acceptors (B, Al, Ga, In), group IV semiconductors (C, Si, Ge), and group V donors (P, As, Sb). III–V compounds pair the two outer columns. Textbook figure © McGraw-Hill; reproduced from S. Levkov, NJIT ECE 271 lecture slides, Ch. 2.*

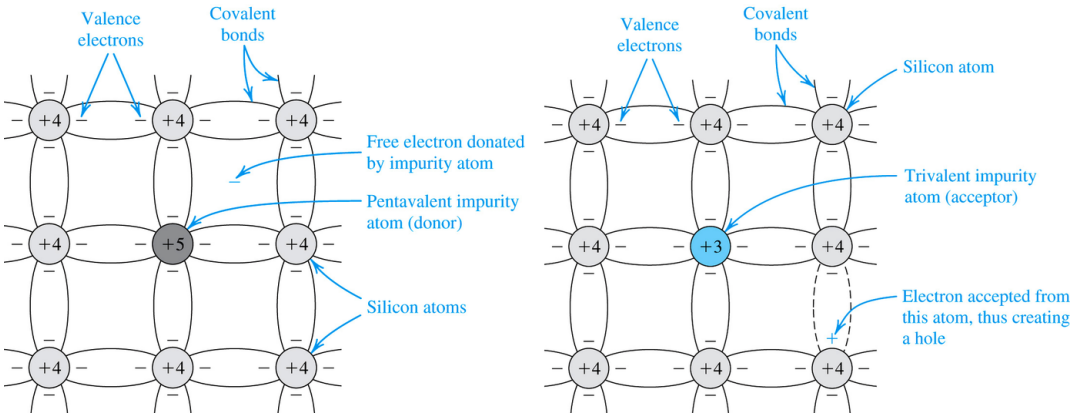


Figure 2.4. *Doping in the bond picture. Left: a group V donor has a fifth electron that is not needed for bonding and is easily freed. Right: a group III acceptor leaves one bond incomplete; an electron from a neighbouring bond can fill it, and the vacancy that moves is a hole. Textbook figures © McGraw-Hill; reproduced from S. Levkov, NJIT ECE 271 lecture slides, Ch. 2.*

1.3 Where Boltzmann statistics stop working

Modern source/drain regions are doped above 10^{20} cm^{-3}, which pushes E_F into the conduction band. The material is then **degenerate**, the Boltzmann approximation fails, and the full Fermi–Dirac integral is required. Two consequences matter here: the mass-action law in the form $np = n_i^2$ no longer holds in those regions, and bandgap narrowing of order 100 meV sets in. Neither affects the channel, but both affect source resistance and injection — and Section 4.1 will argue that source injection is now the thing that limits drive current.

2 Transport

2.1 Scattering, mean free time and mean free path

In equilibrium carriers are not stationary. They move with a thermal velocity

$$v_{th} = \sqrt{\frac{3kT}{m^*}} \approx 2.3 \times 10^7 \text{ cm/s}$$

for electrons in silicon at 300 K, in random directions, so the net current is zero. They are interrupted by collisions — with lattice vibrations (phonons), with ionised dopants, with interfaces — at a mean interval τ, travelling a **mean free path** $\lambda = v_{th}\tau$ between them.

Keep λ in view. It is the number that decides whether anything else in this section is applicable, and Section 4.1 returns to it.

2.2 Mobility

Apply a field. Between collisions a carrier accelerates; at each collision its momentum is randomised. Averaging over the ensemble gives a steady drift superimposed on the thermal motion:

$$v_d = -\mu_n \mathcal{E} \quad \text{(electrons)}, \qquad \boxed{\mu = \frac{q\tau}{m^*}}$$

Mobility is therefore large when collisions are rare (τ large) and when the carrier is light (m^* small). Those are the only two levers, and every mobility engineering technique in this module pulls one of them.

> **Common pitfall.** Some derivations give $\mu = q\tau/2m^*$, with a factor of two. It comes from assuming every free flight lasts exactly τ, so the velocity ramps from zero to $q\mathcal{E}\tau/m^*$ and averages half that. But collisions are random events, so free times are *exponentially distributed* with mean τ. Long flights both last longer and reach higher velocity, so they carry more weight in the time average; with $\langle t^2 \rangle = 2\tau^2$ the average drift velocity becomes $q\mathcal{E}\langle t^2 \rangle / (2m^* \langle t \rangle) = q\mathcal{E}\tau/m^*$. Use $\mu = q\tau/m^*$.

2.3 Mobility depends on doping and temperature

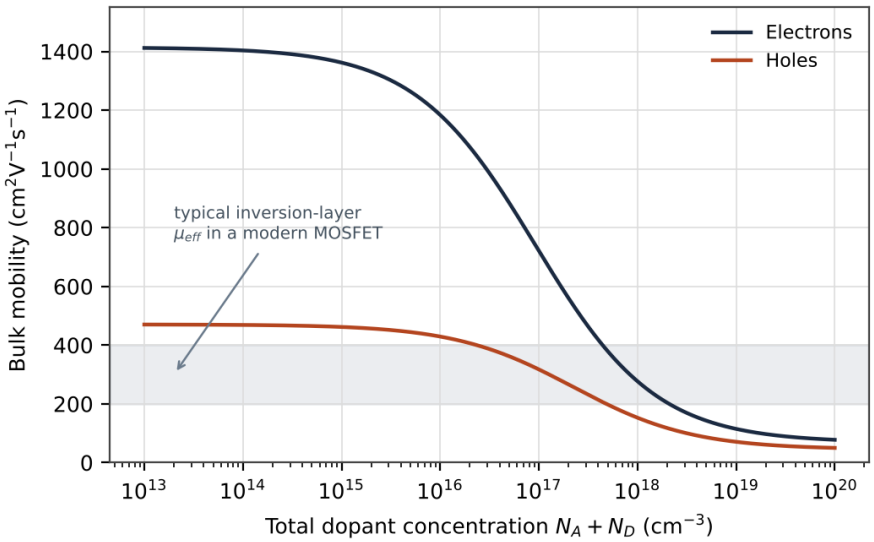


Figure 2.5. *Bulk mobility in silicon at 300 K as a function of total dopant concentration, computed from the Caughey–Thomas empirical model with standard parameters. At low doping, phonon scattering dominates and mobility is doping-independent; above roughly* 10^{16} cm^{-3}*, ionised-impurity scattering takes over. The shaded band marks the effective mobility actually found in a modern inversion layer — see Section 4.3 for why it is so much lower.*

The two regimes have opposite temperature dependence, which is a standard exam question. Phonon scattering gets worse as the lattice vibrates harder, so $\mu \downarrow$ as $T \uparrow$. Ionised-impurity scattering gets *weaker* as carriers move faster past the dopant and spend less time being deflected, so $\mu \uparrow$ as $T \uparrow$. Lightly doped material therefore loses mobility on heating; heavily doped material gains it.

2.4 High fields: velocity saturation

The linear relation $v = \mu\mathcal{E}$ is a low-field approximation. Once a carrier gains enough energy between collisions to emit an optical phonon, that emission channel opens and removes energy as fast as the field supplies it. The velocity stops increasing.

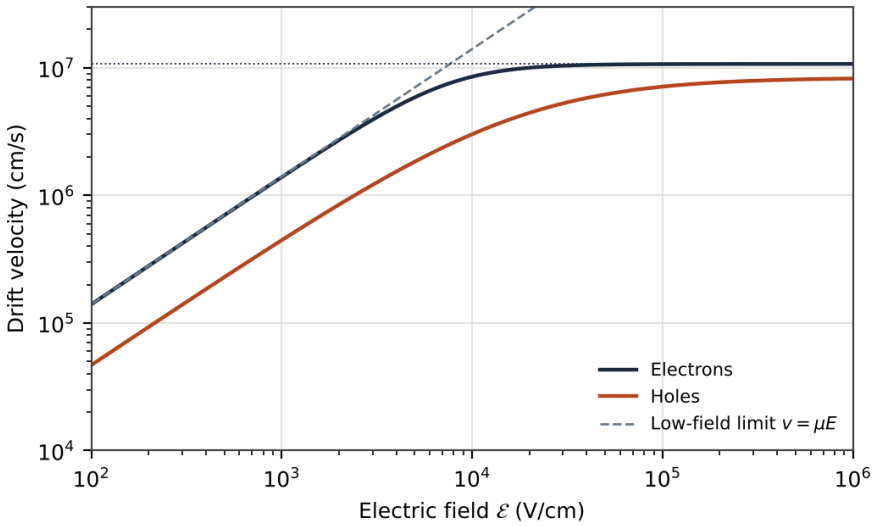


Figure 2.6. *Drift velocity against electric field in silicon at 300 K (Caughey–Thomas form, Canali saturation velocities). The dashed line is the low-field relation $v = \mu\mathcal{E}$, which is already significantly in error by 10^4 V/cm. For a 0.7 V supply across a 15 nm channel the average field is of order 5×10^5 V/cm — well into saturation.*

$$v = \frac{\mu\mathcal{E}}{[1 + (\mu\mathcal{E}/v_{sat})^{\beta}]^{1/\beta}}, \qquad v_{sat} \approx 1.0 \times 10^7 \text{ cm/s (electrons, 300 K)}$$

The last sentence of the caption is the point. **A modern transistor does not operate in the low-field regime at all.** It operates in saturation, which is why the square-law model of Section 3.4 overestimates drive current in real devices, and why Document 3 treats velocity saturation as a short-channel effect rather than a curiosity.

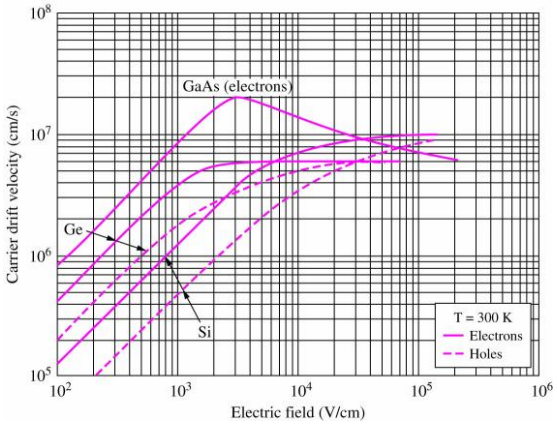


Figure 2.7. *Measured drift velocity against field for electrons and holes in Ge, Si and GaAs at 300 K. Electrons in GaAs overshoot and then fall back as they transfer to heavier conduction-band valleys — the negative differential mobility behind the Gunn effect. Textbook figure © McGraw-Hill; reproduced from S. Levkov, NJIT ECE 271 lecture slides, Ch. 2.*

2.5 Diffusion and the Einstein relation

A concentration gradient drives a flux even with no field. Fick's first law gives, for the current densities,

$$J_n^{\text{diff}} = qD_n \frac{dn}{dx}, \qquad J_p^{\text{diff}} = -qD_p \frac{dp}{dx}.$$

Drift and diffusion are consequences of the *same* collisions, so D and μ are not independent. In non-degenerate material,

$$\frac{D_n}{\mu_n} = \frac{D_p}{\mu_p} = \frac{kT}{q} = U_T.$$

At 300 K, $U_T = 25.9$ mV. (Many texts quote 25 mV, which is the value at 290 K.)

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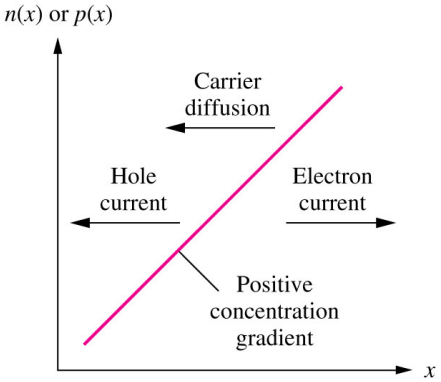


Figure 2.8. *Diffusion runs down a concentration gradient, for holes and electrons alike. Because the electron charge is negative, electron diffusion current flows opposite to the electron flux, while hole current flows with it. Textbook figure © McGraw-Hill; reproduced from S. Levkov, NJIT ECE 271 lecture slides, Ch. 2.*

2.6 The drift-diffusion equation, and what it assumes

Summing both mechanisms:

$$J_n = q\mu_n n\mathcal{E} + qD_n \frac{dn}{dx}, \qquad J_p = q\mu_p p\mathcal{E} - qD_p \frac{dp}{dx}.$$

This is the workhorse of classical device physics and of every TCAD simulator's default model. Before using it, note what had to be true to derive it:

1. **Many collisions.** Carriers must scatter often enough on the length scale of interest for a drift *velocity* — an ensemble average — to be meaningful. This requires $\lambda \ll L$.
2. **Local equilibrium.** The carrier distribution at each point must be characterised by the local field and the lattice temperature, which requires the potential to vary slowly over λ.
3. **Classical particles.** No quantisation of the carrier's energy levels, and no tunnelling.
4. **Non-degenerate statistics**, for the Einstein relation in the form above.

Every one of these four assumptions is violated somewhere in a modern transistor. Hold that thought until Section 4.

3 The MOSFET

This section builds the device. Everything in Document 3 is a perturbation on the long-channel MOSFET constructed here.

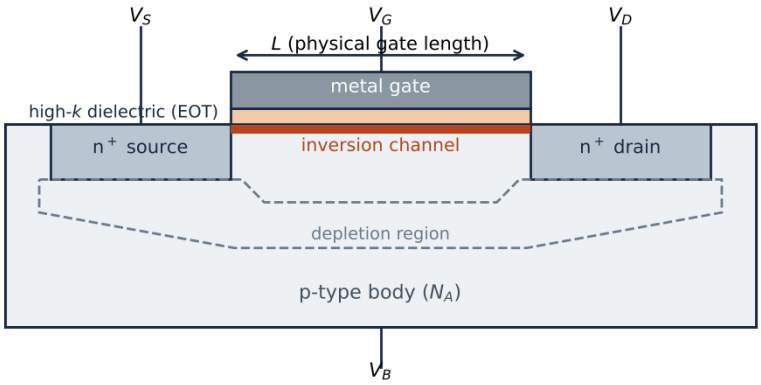


Figure 2.9. *Cross-section of an n-channel MOSFET, with the terminology used throughout this module and Document 3. Note that L denotes the physical gate length, not the technology node name.*

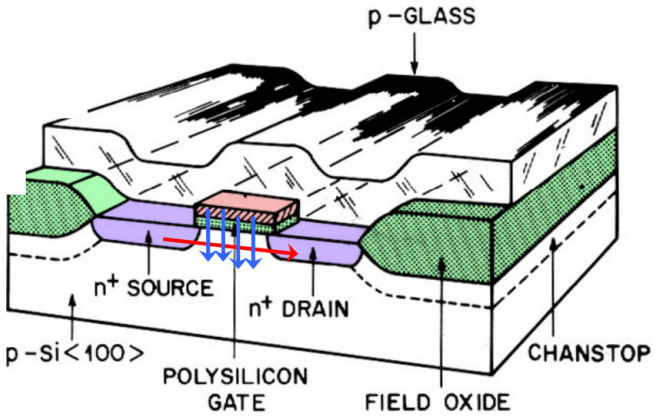


Figure 2.10. *A classic perspective drawing of a planar MOSFET from the LOCOS era: polysilicon gate, n^+ source and drain, field oxide and channel-stop implant isolating neighbouring devices. Original source not stated; reproduced from W. A. Doolittle, Georgia Tech lecture slides (2011).*

3.1 The MOS capacitor: three regimes

Everything the gate does, it does through the MOS capacitor formed by the gate, the dielectric and the body. Consider a p-type body and sweep the gate voltage.

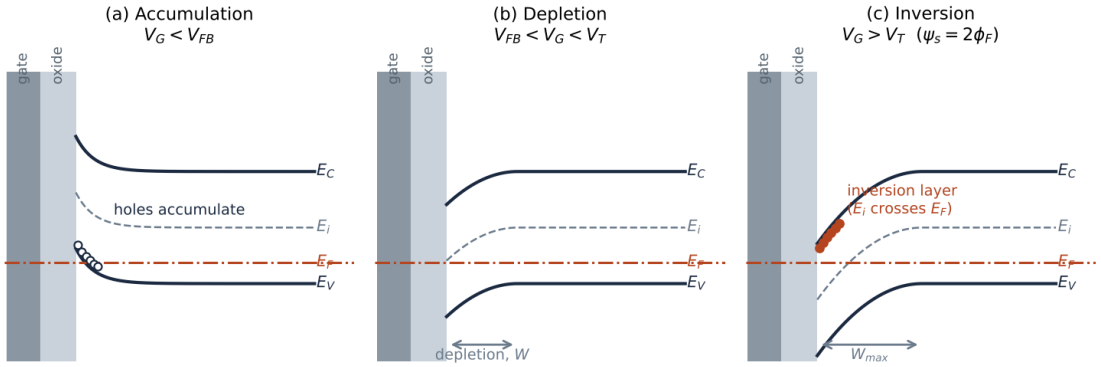


Figure 2.11. *Band diagrams of a MOS capacitor on a p-type body. The oxide and gate are shown as blocks; only the semiconductor bands are drawn, since that is where the physics of interest happens. E_F is flat because the structure is in equilibrium — no current flows through the dielectric.*

Accumulation ($V_G < V_{FB}$). The gate is negative; it attracts majority holes to the surface. Bands bend *upward*, E_V approaches E_F at the surface, and hole density there exceeds the bulk value. No channel.

Depletion ($V_{FB} < V_G < V_T$). The gate is positive; it repels holes, leaving behind a region of uncompensated, immobile acceptor ions. Bands bend *downward*. The depletion width grows as

$$W = \sqrt{\frac{2\varepsilon_{Si}\,\psi_s}{qN_A}}.$$

Inversion ($V_G > V_T$). Bend the bands far enough and E_i at the surface drops below E_F: the surface is now *more n-type than it is p-type*, even though the bulk is p-doped. A thin sheet of mobile electrons forms at the interface. That sheet is the channel.

The conventional definition of "far enough" is **strong inversion**, taken as

$$\psi_s = 2\phi_F, \qquad \phi_F = U_T \ln\frac{N_A}{n_i},$$

at which point the surface electron density equals the bulk hole density. Once strong inversion is reached, further gate voltage goes almost entirely into inversion charge rather than into widening the depletion region, so W pins at

$$W_{max} = \sqrt{\frac{4\varepsilon_{Si}\,\phi_F}{qN_A}}.$$

Note that $\psi_s = 2\phi_F$ is a *convention*, not a physical discontinuity. Inversion charge exists below it, falling exponentially. That exponential tail is the subject of Section 3.5, and it is what ended Dennard scaling.

3.2 Threshold voltage

The gate voltage must cover the flat-band voltage, bend the bands by $2\phi_F$, and supply the charge that terminates the depletion-region field:

$$\boxed{V_T = V_{FB} + 2\phi_F + \frac{\sqrt{2q\varepsilon_{Si}N_A(2\phi_F)}}{C_{ox}}}$$

where $V_{FB} = \phi_{MS} - Q_{ox}/C_{ox}$ accounts for the gate–body work function difference and any fixed charge in the dielectric.

Three observations that Document 3 will need:

- V_T rises with N_A — both through ϕ_F and through the square root.
- V_T falls as C_{ox} rises, i.e. as the dielectric gets electrically thinner.
- **The entire depletion charge under the gate is assumed to be supported by the gate.** In a short-channel device it is not: the source and drain junctions support part of it. That single failed assumption is the origin of threshold roll-off, and Document 3 derives the correction.

A modern device sets V_T principally by **metal gate work-function engineering** rather than by channel doping, because channel doping is now undesirable for the variability reasons in Document 3.

3.3 Gate capacitance and EOT

$$C_{ox} = \frac{\varepsilon_0 k}{t_{phys}} = \frac{\varepsilon_0 k_{\mathrm{SiO_2}}}{\mathrm{EOT}}$$

Equivalent oxide thickness expresses the physical stack as the thickness of SiO_2 that would give the same capacitance. For EOT = 1 nm, $C_{ox} = 3.45 \times 10^{-6}\ \mathrm{F/cm^2} = 34.5\ \mathrm{fF}/\mu\mathrm{m}^2$.

Section 4.2 shows that the *electrically effective* EOT is always worse than this, by a quantum-mechanical amount that cannot be engineered away.

3.4 Long-channel current–voltage

Take the **gradual channel approximation**: assume the field along the channel is much weaker than the field across it, so the inversion charge at position y is set by the local vertical bias alone,

$$Q_{inv}(y) = -C_{ox}[V_{GS} - V_T - V(y)].$$

Integrating the drift current from source to drain gives the square-law model:

$$I_D = \mu_{eff} C_{ox} \frac{W}{L}\left[(V_{GS} - V_T)V_{DS} - \frac{V_{DS}^2}{2}\right] \qquad (V_{DS} < V_{GS} - V_T)$$

$$I_D = \frac{\mu_{eff} C_{ox}}{2}\frac{W}{L}(V_{GS} - V_T)^2 \qquad (V_{DS} \geq V_{GS} - V_T)$$

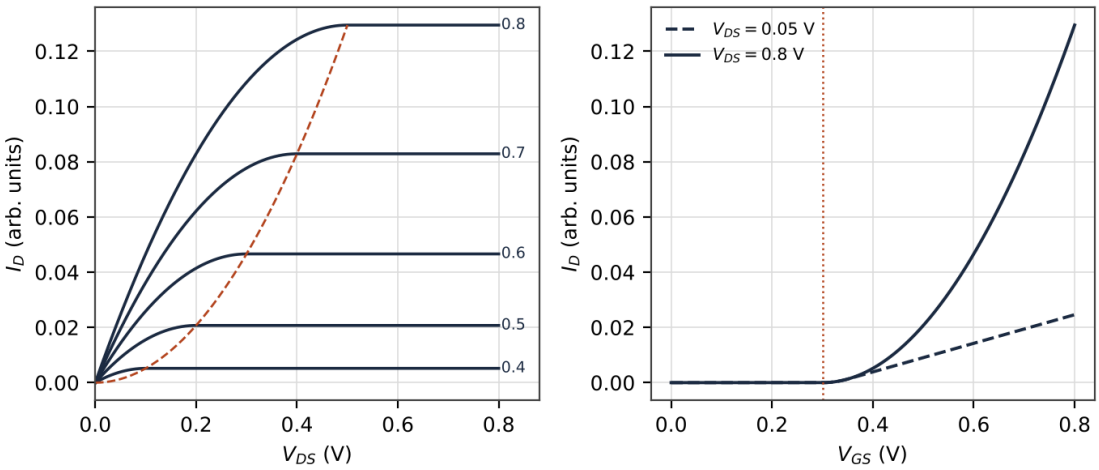


Figure 2.12. *Long-channel MOSFET characteristics computed from the square-law model with EOT = 1 nm, $\mu_{eff} = 300\ cm^2V^{-1}s^{-1}$, $V_T = 0.3$ V. Left: output characteristics, with the dashed locus marking pinch-off. Right: transfer characteristics on a linear scale, showing the square-law dependence in saturation and the apparent sharp turn-on at V_T. Figure 2.13 shows what that "sharp" turn-on actually looks like on a log scale.*

Saturation occurs at $V_{DS} = V_{GS} - V_T$, where the inversion charge at the drain end reaches zero — **pinch-off**. Beyond that point, in the long-channel picture, current is independent of V_{DS}.

Transconductance in saturation follows by differentiation:

$$g_m = \frac{\partial I_D}{\partial V_{GS}} = \mu_{eff} C_{ox} \frac{W}{L} (V_{GS} - V_T).$$

> **Two warnings before you use this model.** First, it assumes constant mobility, which Section 2.4 has already shown to be false at the fields inside a modern device; with full velocity saturation the dependence becomes approximately *linear* in overdrive, $I_D \approx WC_{ox}v_{sat}(V_{GS} - V_T)$, and $g_m \approx WC_{ox}v_{sat}$ — independent of L. Second, the gradual channel approximation fails precisely when the channel is short, which is the subject of Document 3. The square-law model is a scaffold for understanding, not a design tool at these dimensions.

3.5 Subthreshold conduction and the 60 mV/decade limit

This is the most important subsection in Lecture 1.

Below threshold the channel is not off. Inversion charge falls exponentially with surface potential, and the drain current with it. In weak inversion the transport is **diffusive**, and

$$I_D \propto \exp\left(\frac{V_{GS} - V_T}{nU_T}\right).$$

Define the **subthreshold swing** SS as the gate voltage change needed to move the current by one decade:

$$SS = \frac{dV_{GS}}{d(\log_{10} I_D)} = n \frac{kT}{q} \ln 10 = \left(1 + \frac{C_d}{C_{ox}}\right) \frac{kT}{q} \ln 10$$

The body factor $n = 1 + C_d/C_{ox}$ is a **capacitive divider**: the gate competes with the depletion capacitance for control of the surface potential. Perfect control means $C_d \to 0$, so $n \to 1$, so

$$SS_{\min} = \frac{kT}{q}\ln 10 = 0.0259 \times 2.303 = 59.5 \text{ mV/decade} \quad \text{at 300 K.}$$

That limit is thermodynamic, not technological. It comes from the Boltzmann tail of the carrier energy distribution. No amount of gate engineering beats it, because the barrier the gate modulates is being crossed by carriers whose energy distribution is set by temperature alone. You can approach 60 mV/dec; you cannot go below it with a thermionic device.

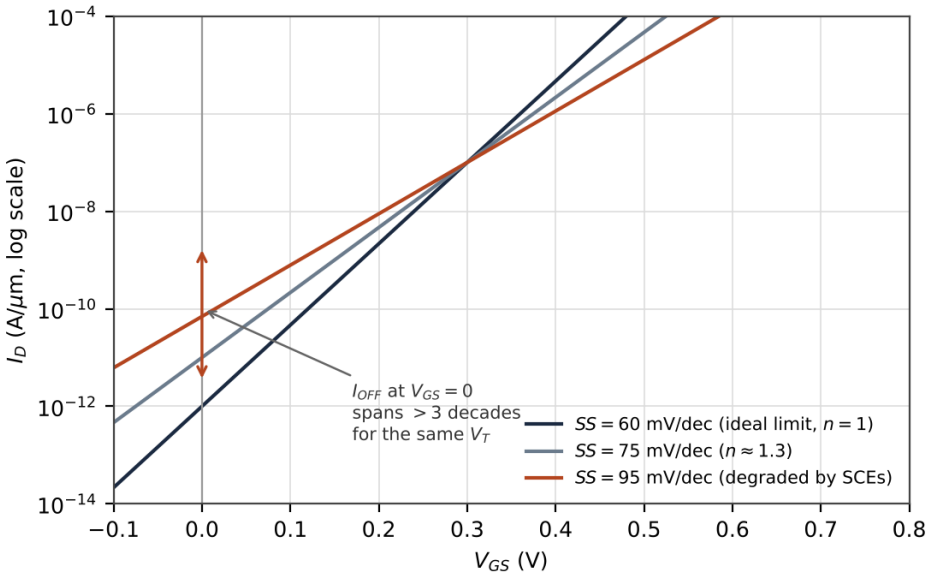


Figure 2.13. *Subthreshold characteristics for the same threshold voltage but three different swings. The threshold voltage is identical in all three cases; only the steepness differs. The off-state current at $V_{GS} = 0$ spans more than three orders of magnitude as a result.*

Why this ended Dennard scaling

Now connect it to Document 1. Off-state leakage per device scales as

$$I_{OFF} \propto 10^{-V_T/SS}.$$

With $SS \approx$ 60–100 mV/dec, **every** 60 **to** 100 **mV removed from** V_T **multiplies the leakage of every transistor on the die by ten.** Constant-field scaling demanded exactly that reduction at every node. By the early 2000s, with 10^8 devices per chip, the product of per-device leakage and device count became comparable to switching power. V_T scaling stopped; V_{DD} scaling stopped with it; power density began to rise.

This is the whole chain, in one equation. It is also why the entire industry cares about C_d/C_{ox} — the only term in SS that engineering can touch. Thin-body SOI, FinFET and gate-all-around architectures are, at bottom, attempts to drive C_d toward zero. Document 3 makes that case properly.

And it is why **tunnel FETs, negative-capacitance FETs and other steep-slope devices** are a live research area: they are attempts to sidestep the Boltzmann tail entirely by changing the injection mechanism. A good starting point is A. M. Ionescu and H. Riel,

"Tunnel field-effect transistors as energy-efficient electronic switches", *Nature*, vol. 479, pp. 329–337, 2011.

3.6 Body effect

Reverse-biasing the body relative to the source widens the depletion region and increases the charge the gate must support, raising V_T:

$$V_T(V_{SB}) = V_{T0} + \gamma\left(\sqrt{2\phi_F + V_{SB}} - \sqrt{2\phi_F}\right), \qquad \gamma = \frac{\sqrt{2q\varepsilon_{Si}N_A}}{C_{ox}}.$$

Worth knowing because it is heavily used in circuit design, and worth noting that it largely **disappears** in fully depleted thin-body and gate-all-around devices, where there is no quasi-neutral body left to bias. The disappearance of the body terminal is one of the less-discussed consequences of the FinFET transition.

4 Where this picture breaks

Sections 1–3 are a competent account of a transistor with a channel one micron long. This section is about why that account is, at present dimensions, being applied past its limit. **This is the material that distinguishes this module from an undergraduate device course.**

4.1 $\lambda \approx L$: transport is quasi-ballistic

Take the numbers seriously. For electrons in a silicon inversion layer at 300 K:

$$\tau = \frac{\mu_{eff} m_n^*}{q} = \frac{(0.030\ \mathrm{m^2V^{-1}s^{-1}})(0.26 \times 9.11 \times 10^{-31}\ \mathrm{kg})}{1.602 \times 10^{-19}\ \mathrm{C}} \approx 4.4 \times 10^{-14}\ \mathrm{s} \approx 44\ \mathrm{fs}$$

$$v_{th} = \sqrt{3kT/m_n^*} \approx 2.3 \times 10^7\ \mathrm{cm/s}$$

$$\lambda = v_{th}\tau \approx 1.0 \times 10^{-6}\ \mathrm{cm} = \boxed{\approx 10\ \mathrm{nm}}$$

Now compare with the physical gate length of a transistor in volume production today, which Document 1, Section 7 established is of the order of **12 to 20 nm**.

$$\frac{L}{\lambda} \approx 1 \text{ to } 2.$$

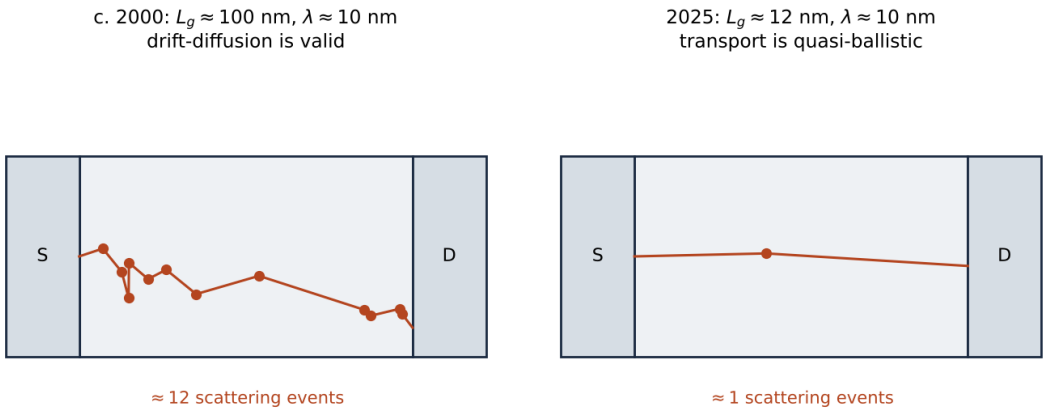


Figure 2.14. *The same mean free path, two different channel lengths. Left: the situation assumed by most undergraduate treatments of the MOSFET, appropriate to a* 100 *nm gate around the year 2000 — a carrier scatters many times crossing the channel, so drift velocity and mobility are meaningful ensemble averages. Right: a channel at current dimensions, where a carrier may scatter once, or not at all. Trajectories are illustrative, generated with exponentially distributed free paths of mean* 10 *nm.*

This invalidates assumption 1 of Section 2.6, and with it assumption 2. In the right-hand case, mobility is not a property of the channel in any clean sense. Drift velocity is not an average over many collisions. The drift-diffusion equation is being extrapolated outside its derivation.

> **Common pitfall.** Many lecture notes compute $\lambda \approx 10$ nm, compare it with a gate length of about 0.1 μm, and conclude that carriers undergo many collisions crossing the device. That was true around the year 2000. The mean free path has not changed — it is set by the material and the scattering — but the gate length has fallen five- to eight-fold. The conclusion is now false.

What replaces it is the **ballistic/quasi-ballistic picture** associated with Natori and with Lundstrom. In the limit, drive current is not set by transport through the channel at all; it is set by the rate at which the source injects carriers over the barrier at the top of the channel:

$$I_{D,\text{ballistic}} = W\, Q_{inv}\, v_{inj},$$

with v_{inj} the thermal injection velocity, of order 10^7 cm/s and *independent of channel length*. Real devices sit between the two limits, usually described by a backscattering coefficient. Three consequences worth stating:

- **Drive current stops improving when you shorten the channel.** Shortening L was the classical route to more current; in the ballistic limit there is nothing left to gain.
- **Source engineering becomes the lever.** If injection sets the current, then source resistance and the source injection velocity are what matters — which is why parasitic resistance is now a dominant design concern.
- **Mobility remains a useful figure of merit, but is no longer a transport coefficient.** It correlates with backscattering, which is why strain engineering still works.

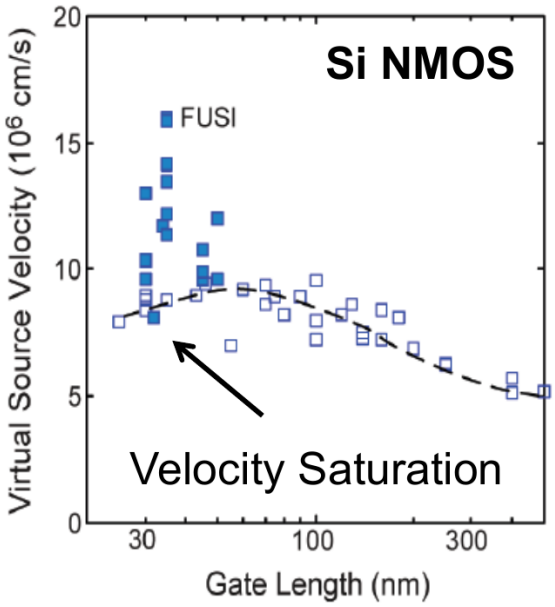


Figure 2.15. *Measured virtual-source (injection) velocity of silicon n-MOSFETs against gate length. It rises only slowly as the gate shrinks and levels off at around* 10^7 *cm/s: the device current is set by injection from the source, not by transit along the channel. Source: A. Khakifirooz and D. A. Antoniadis, IEEE Trans. Electron Devices 55, 1391 (2008); reproduced from ETH Zürich lecture slides.*

4.2 Inversion-layer quantisation

The inversion layer is a potential well perhaps 1–2 nm wide, formed between the dielectric barrier and the band bending. A well that narrow quantises the carrier motion perpendicular to the interface into discrete subbands — carriers are confined in one dimension and free in the plane. This is a **two-dimensional electron gas**, and it is the same physics Lecture 2 develops for quantum wells.

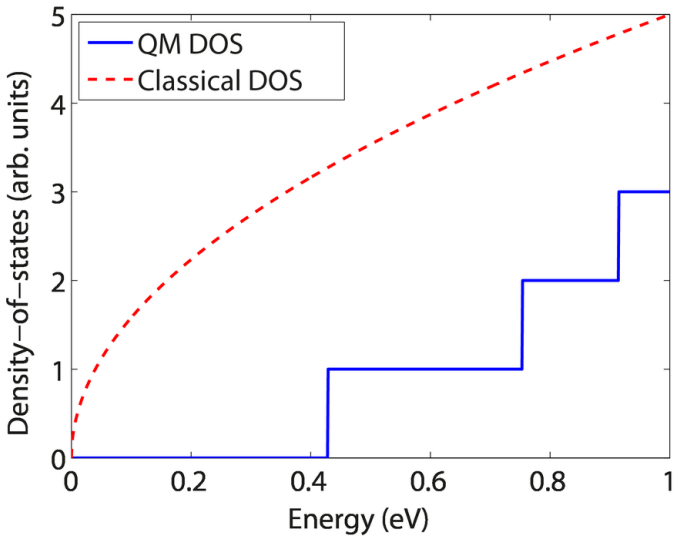


Figure 2.16. *Density of states of carriers confined in a thin channel (solid) against the classical bulk result (dashed). Confinement turns the smooth* $\sqrt{E}$ *curve into a staircase whose first step sits above the band edge. Reproduced from ETH Zürich lecture slides.*

Three consequences, all of which cost performance:

1. **The lowest allowed state is above the classical band edge.** More band bending, and therefore more gate voltage, is needed to populate it: V_T shifts upward, by tens of millivolts up to about 100 mV, depending on doping and field.
2. **The charge centroid sits roughly 1 nm away from the interface**, not at it. That separation adds a capacitance in series with C_{ox}, degrading the electrically effective EOT by about 0.3–0.4 nm. At an EOT of 1 nm that is a 30–40% penalty, and it is irreducible.
3. **The density of states changes**, from the three-dimensional $\sqrt{E}$ form to the two-dimensional staircase, altering the relation between surface potential and charge.

Assumption 3 of Section 2.6 — classical particles — therefore fails too. The quantitative version of Section 3.2 and Section 3.4 for a real device requires a coupled Schrödinger–Poisson solution, which is exactly what commercial TCAD does.

4.3 Effective mobility is not bulk mobility

Figure 2.5 showed bulk mobility reaching 1414 $\mathrm{cm^2V^{-1}s^{-1}}$ at low doping. The effective mobility in a modern inversion layer is typically 200–400. The gap has three causes, and getting the explanation right matters:

- **Surface roughness scattering**, dominant at high vertical field. The Si–dielectric interface is not atomically flat, and carriers pressed against it by the gate field scatter off that roughness.
- **Remote Coulomb and remote phonon scattering** from charges and polar vibrational modes in the high-k dielectric. This was one of the costs of adopting hafnium-based dielectrics.
- **Increased phonon scattering** in the confined 2D system.

Plotted against effective vertical field, data from devices of widely varying doping and oxide thickness collapse onto a single **universal mobility curve** — a useful and genuinely predictive result.

> **Common pitfall.** A widely read online text explains surface mobility degradation by saying the electron wavefunction "extends into the oxide" where mobility is lower. This is not the mechanism. Carriers do not conduct in the oxide; the wavefunction tail into the barrier is evanescent and carries no current. The degradation is scattering at and near the interface, as listed above.

4.4 What survives

It would be wrong to leave the impression that the classical picture is useless. Stated carefully:

Concept	Status at $L \approx 15$ nm
Band structure, effective mass	**Valid.** More necessary than before
Doping, mass action, ϕ_F	**Valid** in the channel; fails in degenerate S/D
Electrostatics, V_T, C_{ox}	**Valid**, with quantum corrections to V_T and EOT
Subthreshold swing, 60 mV/dec	**Valid and binding.** Thermodynamic, not technological

Concept	Status at $L \approx 15$ nm
Mobility as a transport coefficient	**Questionable.** Better read as a backscattering figure of merit
Drift-diffusion as the transport model	**Outside its domain.** Usable with calibration, not from first principles
Square-law I_D model	**Pedagogical only.** Velocity saturation dominates

The electrostatics survive. The transport does not. That asymmetry is worth stating plainly, because it explains the shape of the industry roadmap in Document 1: almost every architectural change of the last fifteen years — thin body, FinFET, gate-all-around — is an *electrostatic* fix, because electrostatics is the part we still control.

5 Summary, and what comes next

What to carry forward:

1. $\mu = q\tau/m^*$. Light carriers and rare collisions. Two levers, nothing else.
2. $U_T = kT/q = 25.9$ mV at 300 K — and V_T is a *different quantity*.
3. Inversion begins when $\psi_s = 2\phi_F$, and V_T is what it costs to get there.
4. $SS = (1 + C_d/C_{ox})(kT/q)\ln 10 \geq 59.5$ mV/decade. Thermodynamic. The only engineering handle is C_d/C_{ox}.
5. $\lambda \approx 10$ nm, $L \approx 12$–20 nm. Transport is quasi-ballistic and drift-diffusion is being extrapolated.
6. The electrostatics survive the shrink; the transport model does not.

Where Document 3 picks up. Everything above assumed that the gate, and only the gate, controls the channel potential. Document 3 removes that assumption. When the channel is short enough that the source and drain depletion regions are a significant fraction of it, the drain begins to compete with the gate for control of the barrier — and the result is threshold roll-off, DIBL, punchthrough, degraded subthreshold swing, and the leakage that follows from them. It will also show that these are not six separate problems but one problem, with one family of solutions, and that those solutions are the devices studied in the rest of this module.

Sources and further reading

Standard texts (either is sufficient; Taur and Ning is the better fit for this module):

- Y. Taur and T. H. Ning, *Fundamentals of Modern VLSI Devices*, 3rd ed., Cambridge University Press, 2021. *Chapters 2–3 cover Sections 1–3 of this document; Chapter 6 covers Section 4.*
- S. M. Sze, Y. Li and K. K. Ng, *Physics of Semiconductor Devices*, 4th ed., Wiley, 2021.

On the material in Section 4, which the textbooks treat only briefly:

- M. Lundstrom, "Elementary scattering theory of the Si MOSFET", *IEEE Electron Device Letters*, vol. 18, no. 7, pp. 361–363, 1997. *The backscattering formulation. Short and readable.*
- K. Natori, "Ballistic metal-oxide-semiconductor field effect transistor", *Journal of Applied Physics*, vol. 76, no. 8, pp. 4879–4890, 1994.

- M. Lundstrom and Z. Ren, "Essential physics of carrier transport in nanoscale MOSFETs", *IEEE Transactions on Electron Devices*, vol. 49, no. 1, pp. 133–141, 2002. *If you read only one paper from this document, read this one.*

Data and models used in the figures:

- Mobility against doping (Fig. 2.5): the empirical form of D. M. Caughey and R. E. Thomas, *Proc. IEEE*, vol. 55, no. 12, pp. 2192–2193, 1967, with widely used silicon fit parameters ($\mu_{max} = 1414$ and 470.5, $\mu_{min} = 68.5$ and 44.9 $\mathrm{cm^2V^{-1}s^{-1}}$, $N_{ref} = 9.2 \times 10^{16}$ and 2.23×10^{17} $\mathrm{cm^{-3}}$, $\alpha = 0.711$ and 0.719, for electrons and holes).
- Velocity against field (Fig. 2.6): Caughey–Thomas form with $\beta_n = 2$, $\beta_p = 1$; saturation velocities 1.07×10^7 cm/s (electrons) and 8.37×10^6 cm/s (holes) from C. Canali *et al.*, *IEEE Trans. Electron Devices*, vol. 22, no. 11, pp. 1045–1047, 1975.
- n_i for silicon: P. P. Altermatt *et al.*, "Reassessment of the intrinsic carrier density in crystalline silicon in view of band-gap narrowing", *J. Appl. Phys.*, vol. 93, no. 3, pp. 1598–1604, 2003.

Figure credits. Figures 2.5, 2.6, 2.12 and 2.13 are computed from the models and parameters cited above. Figures 2.9, 2.11 and 2.14 are schematics drawn for this document; the trajectories in Figure 2.14 are random and change each time it is regenerated. The other figures are reproduced from earlier course material for this module — S. Levkov's NJIT ECE 271 slides (textbook figures © McGraw-Hill), W. A. Doolittle's Georgia Tech slides (2011) and ETH Zürich lecture slides — with the original source, where known, given in each caption. Copyright in reproduced figures remains with the original owners.